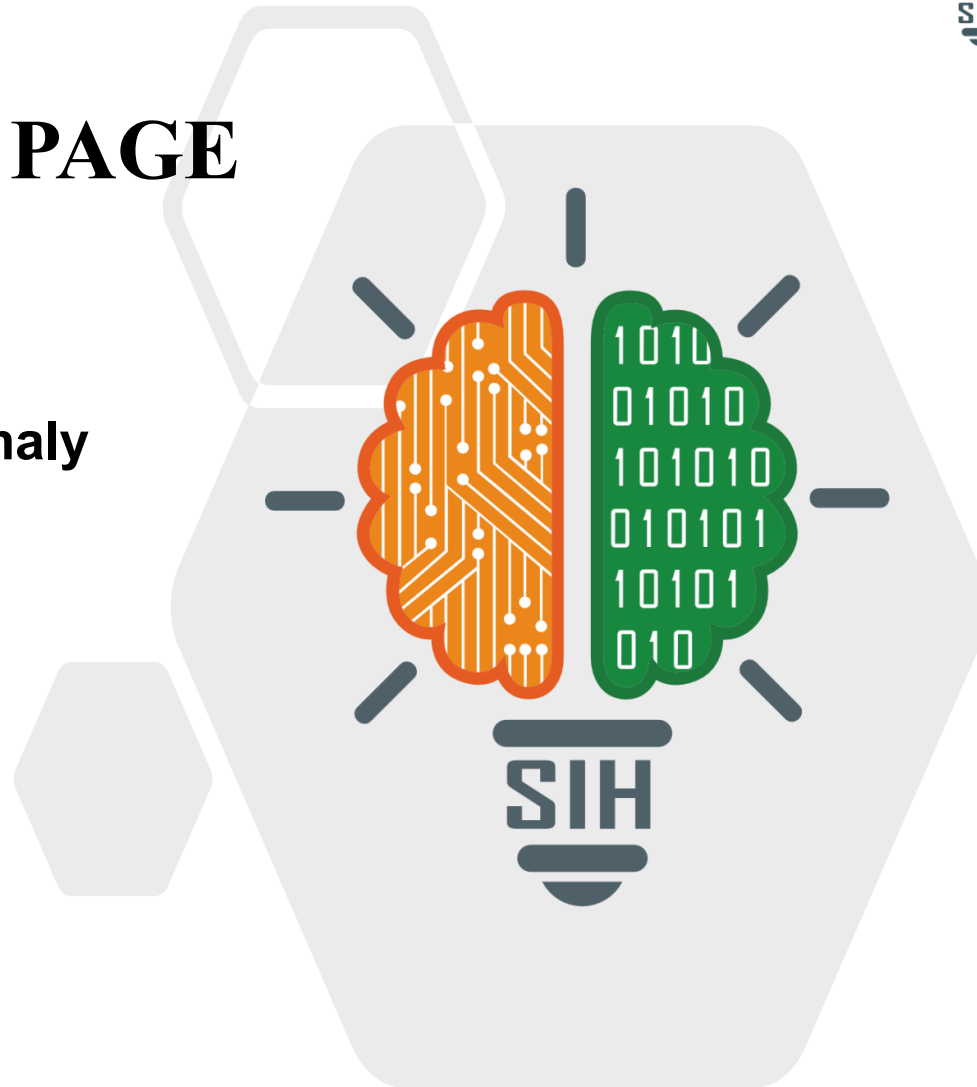


TITLE PAGE

- Problem Statement ID – SIH26170
- Problem Statement Title- AI-Driven Anomaly Detection in Component Burn-In & Screening
- Theme- Smart Automation
- PS Category- Software
- Team ID- (enter SIH portal Team ID)
- Team Name – Pioneers



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- ❖ AETHER: lot-aware PAT outlier screen + 24 h $\rightarrow$ 168 h drift predictor. After 24 h of 125° C burn-in it returns PASS / HOLD / REJECT plus an inspector brief in engineering English.
- ❖ Working prototype: Streamlit QA workstation with Lot board, QA inspector, Screen-my-data, judging rubric. Public demo on Render / Streamlit Cloud.

❖ Detailed explanation of the proposed solution

- ❖ Module A — dynamic outliers: robust PAT ($\text{median} + 6 \times 1.4826 \times \text{MAD}$, AEC-Q001), Isolation Forest, robust Mahalanobis. Scores are lot-normalised so a fast process corner is not scrapped.
- ❖ Module B — drift: Ridge (inspectable) + optional HGB blend forecasts 168 h from 0 h and 24 h only. Flags if predicted slope > healthy 95th percentile or forecast crosses 90% of the datasheet cap.

❖ How it addresses the problem

- ❖ Static 50 μA datasheet PASS on a 45 μA part in a 10 μA lot (SIH example LOTSIH-0045). Dynamic PAT REJECTS it. Latent drift is pulled at 24 h instead of escaping into a payload.

❖ Innovation and uniqueness of the solution

- ❖ Two-signal REJECT (PAT or unsafe drift) — a lone Isolation Forest twitch cannot kill a flight part. HOLD band (not scrap/release). FN cost 1000 vs FP 12. Ridge waterfalls a QA inspector can argue with.
- ❖ Held-out lots (24 lots, 4,945 parts, split by lot): 100% recall (0 miss / 102 latent defects), 24 h REJECT precision 98.7%, IDDQ 168 h MAE 0.51 μA vs 1.26 μA linear extrapolation.

TECHNICAL APPROACH



- **Technologies to be used (programming languages, frameworks, hardware)**
- Python 3.12 · scikit-learn (IsolationForest, MinCovDet, RidgeCV, HistGradientBoosting) · pandas / NumPy · joblib frozen bundle
- Streamlit + Plotly QA workstation · pytest · GitHub Actions-ready scripts (train.py, evaluate.py, predict_realtime_iddq.py)
- No extra hardware: laptop / 512 MB web instance. Optional ATE CSV ingest. Deploy: Render Blueprint + Streamlit Community Cloud
- **Methodology and process for implementation (flow / working prototype)**
- 1. Physics-informed lots (0 / 24 / 96 / 168 h IDDQ, leakage, tpd) — ISRO flight data is not public
- 2. Split by LOT (not by part) → train Module A + Module B on train lots, freeze on val, score unseen test lots
- 3. Inference uses only 0 h and 24 h. 96 h / 168 h stay hidden at the early gate
- 4. Fusion: $\text{outlier_score} = 0.40 \text{ PAT} + 0.35 \text{ Isolation Forest} + 0.25 \text{ Mahalanobis}$ → PASS | HOLD | REJECT
- Flow: Lot CSV → PAT / IF / Mahalanobis + 168 h forecast → decision + traveller brief → QA override
- Prototype live: Screen-my-data tab (upload lot or one part vs a reference lot). Pinned example LOTSIH-0045.

FEASIBILITY AND VIABILITY



- **Analysis of the feasibility of the idea**
 - Software-only. Models already trained; artifacts in data/ and models/. Dashboard opens without a GPU.
 - Fits SIH 36-hour finale: train.py / evaluate.py / Streamlit. Tests encode the official 10 μ A / 45 μ A / 50 μ A example.
 - No ITAR data needed. Production path: ATE CSV / STDF $\rightarrow$ freeze PAT k and safety slope on a qualification lot.
- **Potential challenges and risks**
 - False REJECT if the “lot” mixes process corners (fast + slow wafers labelled as one lot).
 - Isolation Forest contamination and PAT k = 6 are policy knobs — a cleaner line may want them tighter or looser.
 - Unit mix-up (nA vs μ A) or swapped 0 h / 24 h columns looks like a lot excursion.
- **Strategies for overcoming these challenges**
 - Lot identity must be real. Two-signal REJECT. QA lead can force HOLD — software is not the last word on a Class-1 part.
 - Unit / serial checks before scoring. Retrain on a known-good historical set when labels are absent.
 - HOLD at 96 h instead of scrap/release so expensive space-grade silicon is not thrown away on a borderline score.

IMPACT AND BENEFITS



- **Potential impact on the target audience**
 - ISRO / Department of Space screening houses, ATE operators, and QA inspectors who today apply static datasheet limits.
 - Catches latent escapes a 50 μA cap misses (textbook: 45 μA in a 10 μA lot $\rightarrow$ static PASS, dynamic REJECT).
 - Held-out: 0 false negatives / 102 latent defects; 1,172 PASS · 202 HOLD · 75 REJECT; 10,800 chamber-hours recovered from early REJECT.
- **Benefits of the solution (social, economic, environmental)**
 - Mission / social: fewer infant-mortality failures in flight payloads — the FN the problem statement penalises.
 - Economic: 144 h of 125° C soak freed per early REJECT; HOLD protects costly space-grade parts from a two-class scrap policy.
 - Environmental: less wasted high-temperature chamber energy on parts that will fail by 168 h anyway.
 - Explainability (SIH rubric): robust z vs lot, PAT hits, predicted slope vs healthy 95th percentile, Ridge feature waterfalls.

- **Details / Links of the reference and research work**

- [1] SIH26170 — AI-Driven Anomaly Detection in Component Burn-In & Screening, ISRO / DoS. <https://sih2026.vuce.in/ps/SIH26170>
- [2] AEC-Q001 Rev-D, Guidelines for Part Average Testing (robust mean ± 6 robust sigma / DPAT).
- [3] MIL-STD-883 Test Method 1015, Burn-in Test · [4] MIL-PRF-38535 Class Q / V screening (160–240 h at 125° C).
- [5][6] Hawkins, Soden et al. — IDDQ testing of CMOS defects (IEEE Trans. Ind. Electron. 1989; IDDQ Testing of VLSI Circuits, 1992).
- [7] Nigh et al., Failure analysis of timing and IDDq-only failures, SEMATECH, ITC 1998 (12 confirmed-defect ICs used as a real check).
- [8] Liu, Ting, Zhou — Isolation Forest, IEEE ICDM 2008. [9] Rousseeuw & van Driessen — MinCovDet, Technometrics 1999.
- [12] Hoerl & Kennard — Ridge Regression, Technometrics 1970. [13] Friedman — Gradient boosting, Ann. Statist. 2001.
- [14] Pedregosa et al. — scikit-learn, JMLR 2011. [15] Elkan — Foundations of Cost-Sensitive Learning, IJCAI 2001.
- Full IEEE list: docs/AETHER_SIH26170_References_and_Sources.docx · Repo: github.com/amol16112005/AETHER-SIH26170